Build a Personalized Online Course Recommender System with Machine Learning

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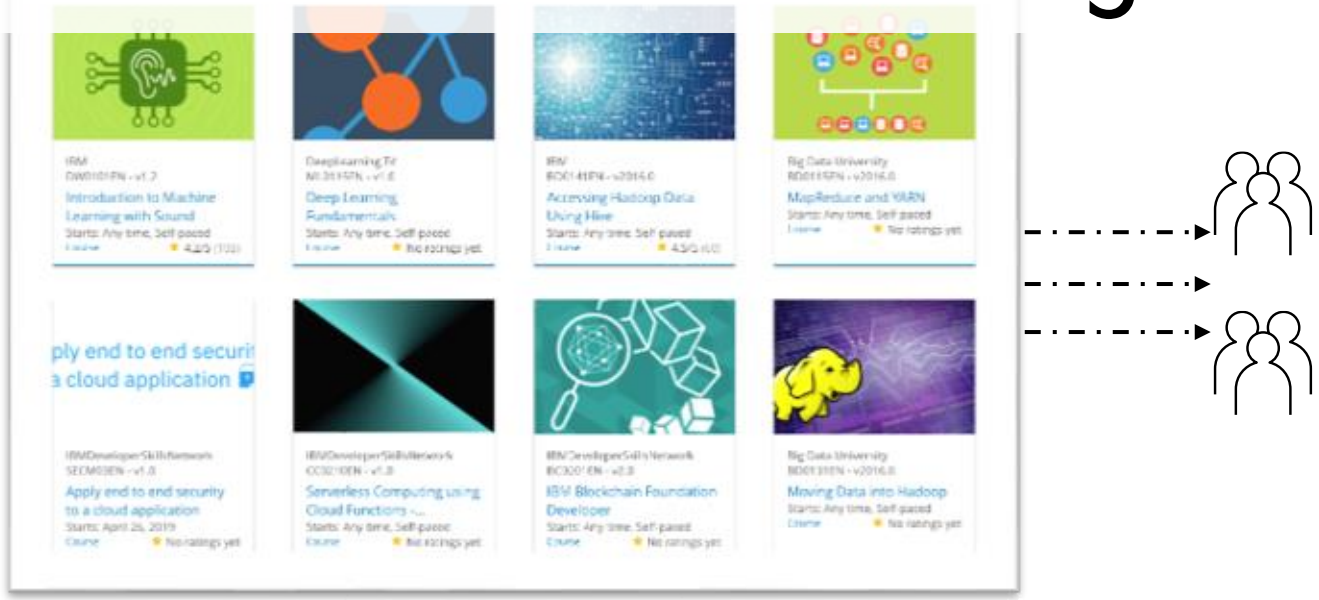
26rd Mar 2026



Outline

- Introduction and Background
- Exploratory Data Analysis
- Content-based Recommender System using Unsupervised Learning
- Collaborative-filtering based Recommender System using Supervised learning
- Conclusion
- Appendix

Introduction

Project Background

- Large-scale **online learning platform** (similar to IBM SkillsBuild / Coursera)
- Dataset contains **33,901 user–course interactions** across **126 courses**
- Project evolution: **Content-Based Filtering** → **Collaborative Filtering** → **Neural Network Embeddings**
- Goal: build a **personalized course recommendation system**

Introduction

Problem Statement

- **Information overload** makes course discovery difficult
- Lack of personalization reduces **engagement and course completion**
- Many relevant courses remain **undiscovered**
- Challenge: model **user preferences** and **course characteristics** to predict ratings

Introduction

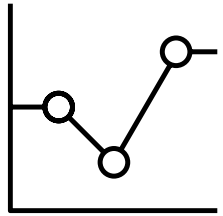
Hypotheses

•**H1 — Latent Space:** Users and courses can be embedded in a shared space where closer distance implies stronger preference

•**H2 — Interaction Modeling:** Combining user & course embeddings enables prediction of user ratings

•**H3 — Complex models :** (Neural Networks or Non-linear Classifiers) will outperform simpler linear baselines by capturing latent user-item relationships.

Exploratory Data Analysis



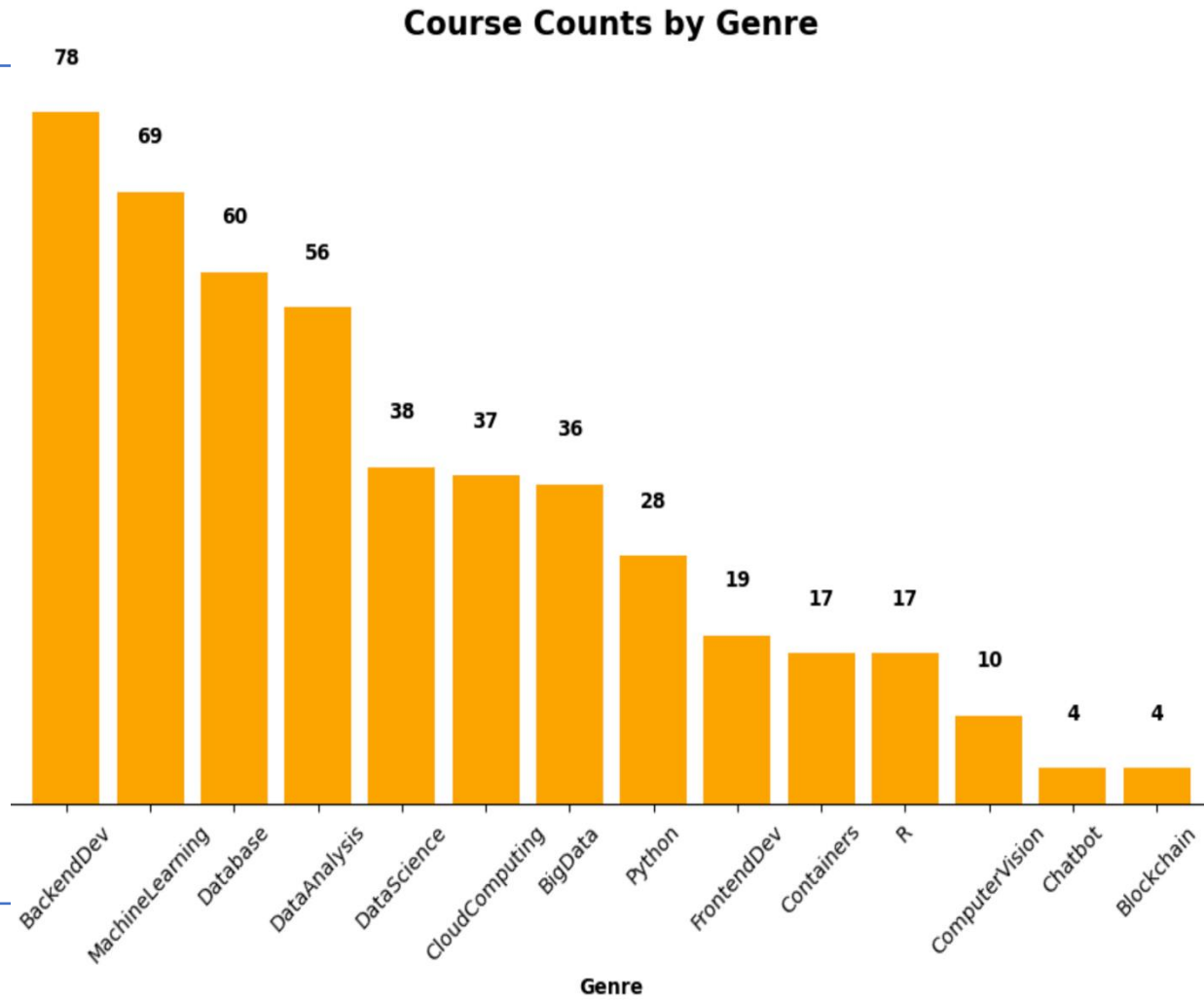
Course counts per genre

Key Insights

- Content concentrated in **technical domains**
- Top genres: Backend Development (78), Machine Learning (69), Database (60)
- Strong focus on Data Science and backend skills

Modeling Impact

- High-volume genres → stronger embeddings
- Low-volume genres → limited interaction learning
- Better recommendation performance in popular categories



Course enrollment distribution

Key Insights

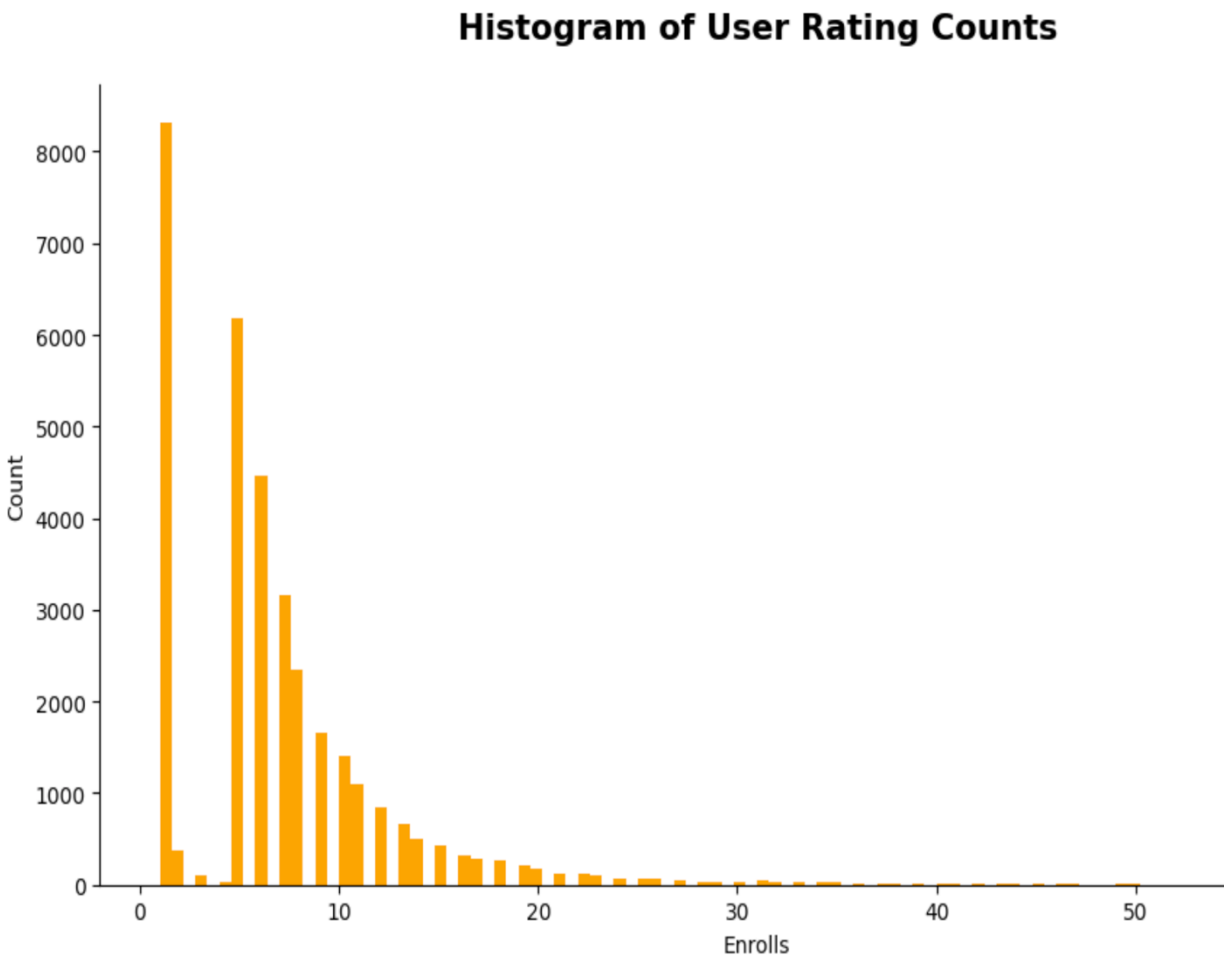
- Distribution follows **long-tail / power law**
- Most users enroll in **1–10 courses**
- Very few **power users** enroll in 40+ courses

User Behavior

- Peak at **1–2 enrollments (~8,000 users)**
- Indicates many **casual or new learners**

Modeling Impact

- Data is **sparse** (few interactions per user)
- Creates **cold start problem** for new users
- Embeddings help by finding **similar users** for recommendations



20 most popular courses

Key Insights

- Foundation courses dominate engagement
- Top courses: **Python for Data Science**, **Introduction to Data Science**
- These courses account for a large share of total interactions

Learning Path Trend

- Users follow a progression:

Foundations → Data Engineering → Advanced Topics

- Example: Python → Big Data → Deep Learning

Modeling Impact

- Top courses have **thousands of ratings**
- Enables **high-quality course embeddings**
- Improves recommendation accuracy

	TITLE	Ratings
0	python for data science	14936
1	introduction to data science	14477
2	big data 101	13291
3	hadoop 101	10599
4	data analysis with python	8303
5	data science methodology	7719
6	machine learning with python	7644
7	spark fundamentals i	7551
8	data science hands on with open source tools	7199
9	blockchain essentials	6719
10	data visualization with python	6709
11	deep learning 101	6323
12	build your own chatbot	5512
13	r for data science	5237
14	statistics 101	5015
15	introduction to cloud	4983
16	docker essentials a developer introduction	4480
17	sql and relational databases 101	3697
18	mapreduce and yarn	3670
19	data privacy fundamentals	3624

Word cloud of course titles

Key Insights

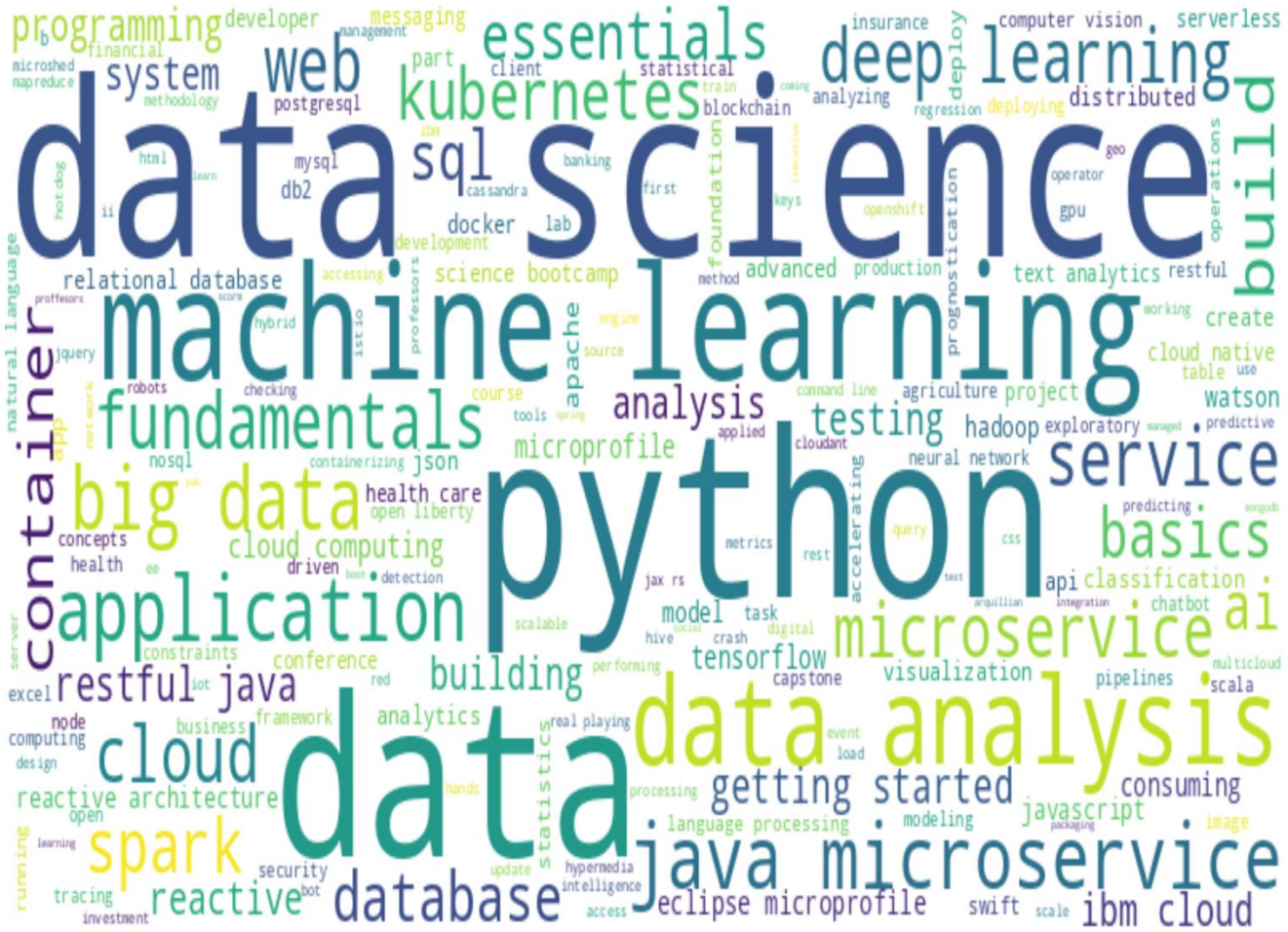
- Dominant keywords: **Data, Science, Python, Machine Learning**
- Confirms strong focus on **AI and data analytics**

Technical Stack

- Infrastructure terms visible: **Cloud, Kubernetes, Spark, Microservices**
- Indicates emphasis on **data engineering & deployment skills**

Modeling Impact

- Keywords define **course similarity**
- Help embeddings learn relationships (e.g., Python ↔ Data Analysis)


data science
machine learning
python
data analysis
data
fundamentals
big data
application
cloud
spark
microservice
java microservice
deep learning
essentials
kubernetes
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system
sql
container
service
basics
ai
build
programming
developer
messaging
relational database
science bootcamp
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text analytics
restful
create
cloud native
watson
analysis
testing
hadoop
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microprofile
neural network
health care
cloud computing
concepts
health
driven
detection
model
task
tensorflow
capstone
visualization
pipelines
scala
classification
chatbot
restful
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building
conference
constraints
analytics
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framework
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natural language
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learning
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packaging
deploy
prognostication
foundation
operations
apache
accelerating
microservice
capstone
visualization
pipelines
scala
multicloud
chatbot
integration
arquillian
jax rs
metrics
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query
scalable
hive
crash
digital
performing
red
iot
excel
real playing
event
load
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ee
boot
detection
driven
health
concepts
health care
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language processing
modeling
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eclipse microprofile
swift
scale
ibm cloud
consuming
javascript
image
packaging
deploy
prognostication
foundation
operations
apache
accelerating
microservice
capstone
visualization
pipelines
scala
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rest
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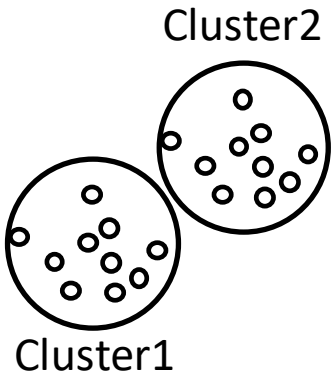
Content-based Recommender System using Unsupervised Learning



Content-Based Recommender System

Recommends courses based on user profile and course genres.

User Profile Vector

User 1078030's profile vector

User Profile Vector		
user1	Python	...
user1	1.0	0

User 1078030's profile vector

Course Genre Vector	
Genre	...
Python	1
Machine Learning	

Course 5's genre vector

2 Calculate Score

Dot product of profile vector and course genre vector

$$\text{Similarity Score} = \text{Dot}(\mathbf{U}, \mathbf{C}_{genre})$$

DOT PRODUCT → Similarity Score

Threshold Check

Threshold Check:
Similarity Score >= 40

YES → Recommend

NO → Do Not Recommend

Threshold Check & Recommendation

Unknown courses ranked based on similarity score

Recommended Courses List	
Course A	★★★★
Course B	★★
Course C	...
Course D	...

Example Non-Recommended	
Course D	...
Course E	...

Performance Metric:
~17.3 New Courses per User Recommended

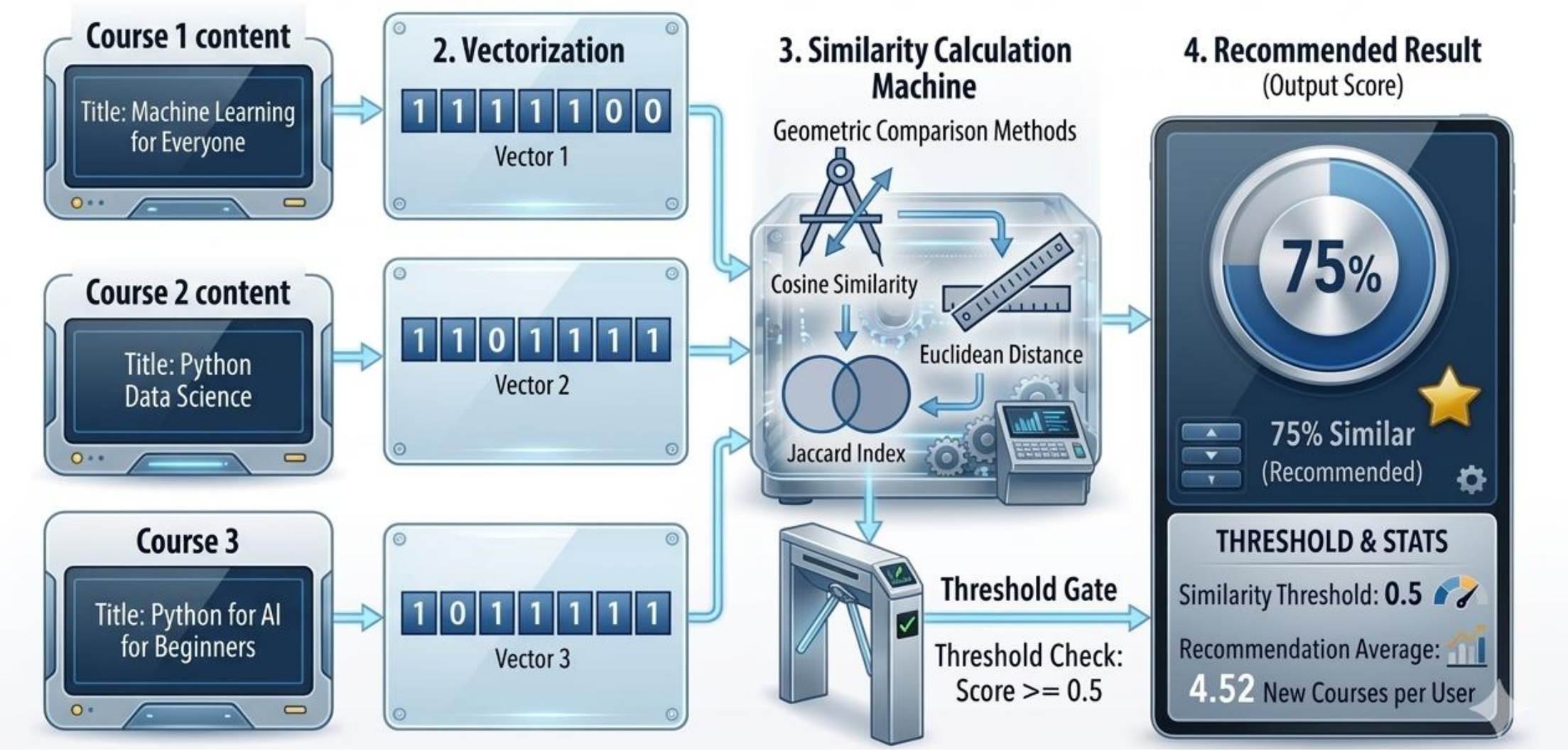
Evaluation results of user profile-based recommender system

- **Operational Threshold:** Recommendations triggered for Interest Score ≥ 40.
- **Recommendation Volume:** ~17.3 new courses discovered per user.
- **System Performance:** Reduces popularity bias; delivers diverse course suggestions.
- **Top Discovery:** excourse72 & excourse73 most recommended, 2,555 times each.

--- Top 10 Most Frequently

	COURSE_ID	Count
0	excourse72	2555
1	excourse73	2555
2	TMP0105EN	2406
3	RP0105EN	2015
4	excourse31	1912
5	SC0103EN	1782
6	GPXX0M6UEN	1634
7	GPXX097UEN	1634
8	excourse03	1634
9	excourse05	1634

Flowchart of content-based recommender system using course similarity



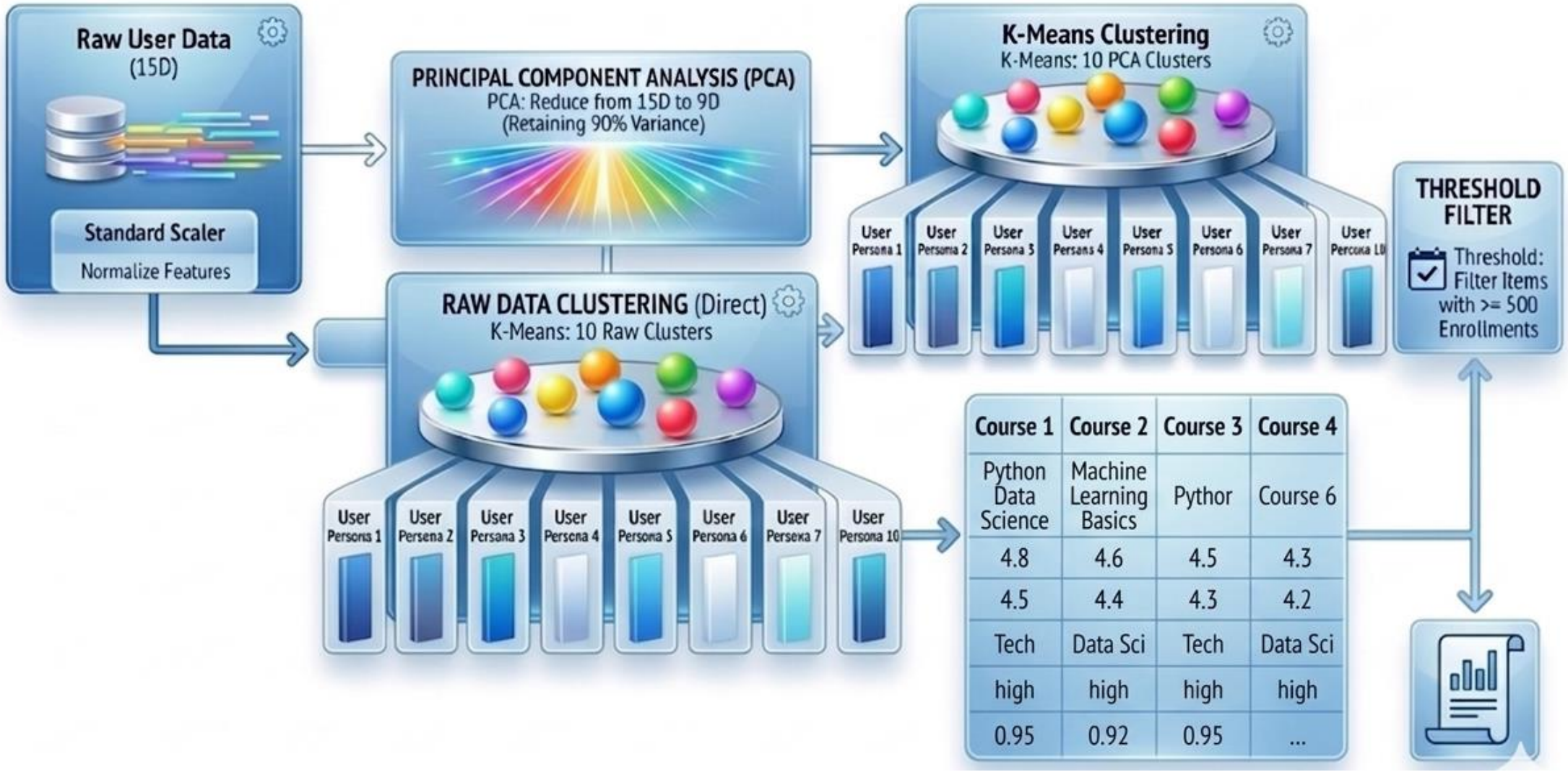
Evaluation results of course similarity-based recommender system

- **Precision Tuning:** 0.5 similarity threshold ensures only close neighbors are recommended.
- **Recommendation Volume:** Avg. **4.52** new courses per user from enrollment history.
- **Top Hub Courses: TMP107** (8,841) and **excourse22** & **excourse62** (7,427) most versatile similar items.

--- Top 10 Most Frequently

	COURSE_ID	Count
0	TMP107	8841
1	excourse22	7427
2	excourse62	7427
3	excourse32	6360
4	DS0110EN	4900
5	excourse68	3423
6	DA0151EN	2812
7	excourse23	2626
8	excourse36	2626
9	excourse65	2540

Flowchart of clustering-based recommender system



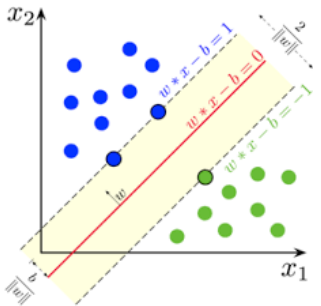
Evaluation results of clustering-based recommender system

- **Operational Threshold:** Minimum **500 enrollments per course within each cluster** to ensure true *group consensus*.
- **Recommendation Volume:** Avg. **10.46 "Next Best Step" courses** suggested per user.
- **Key Course Hub: DS0103EN – Data Science Fundamentals** most recommended (**23,161 counts**).
- **Outcome:** Builds high-confidence **"Discovery Pools"**, reducing choice paralysis using **collective learner behavior**.

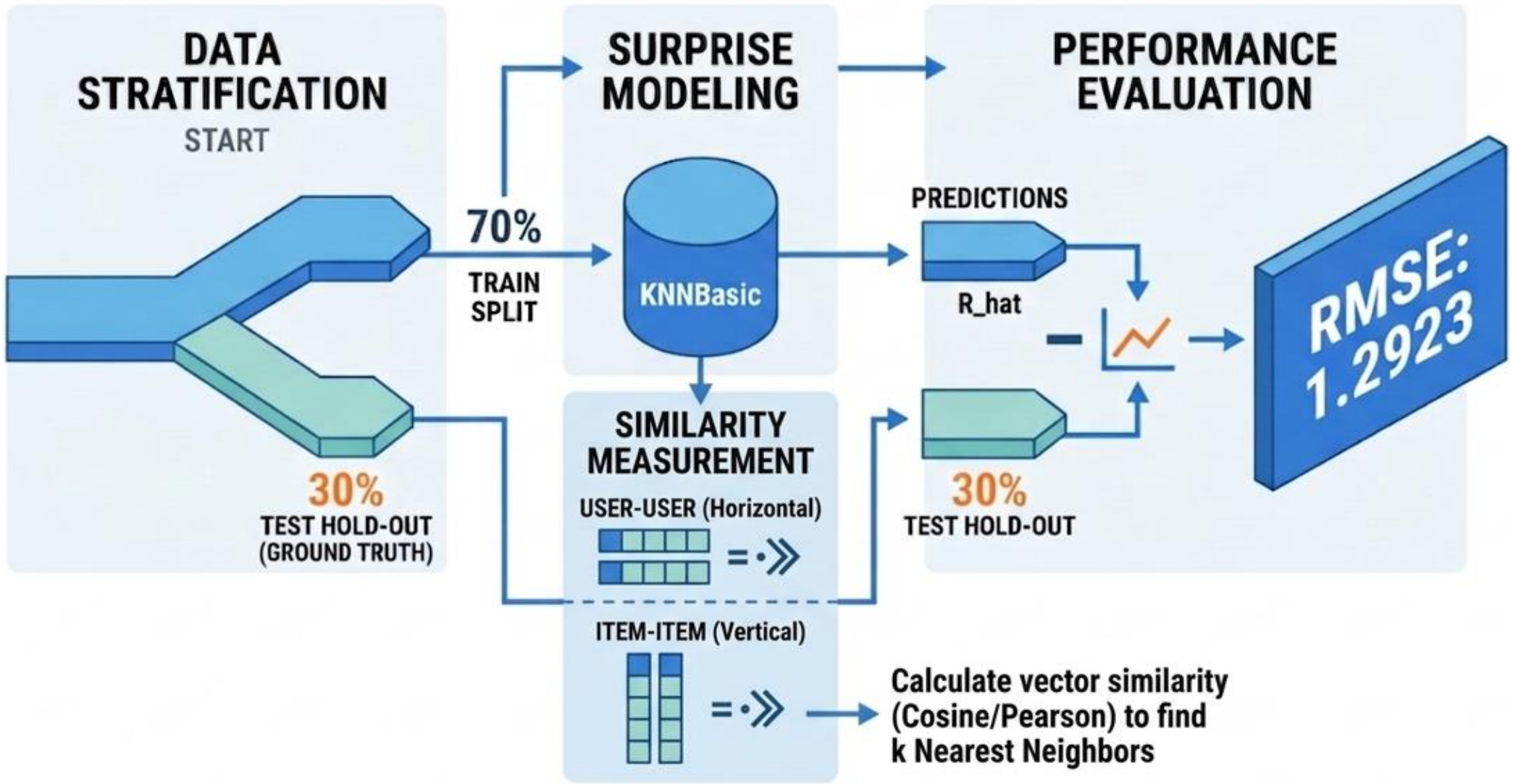
--- Top 10 Most Frequently

	item	Count
0	DS0103EN	23161
1	BD0111EN	22836
2	CC0101EN	21611
3	ST0101EN	21431
4	BD0101EN	20148
5	DS0105EN	19910
6	DS0101EN	18929
7	CL0101EN	14470
8	BC0201EN	11819
9	ML0101ENv3	10962

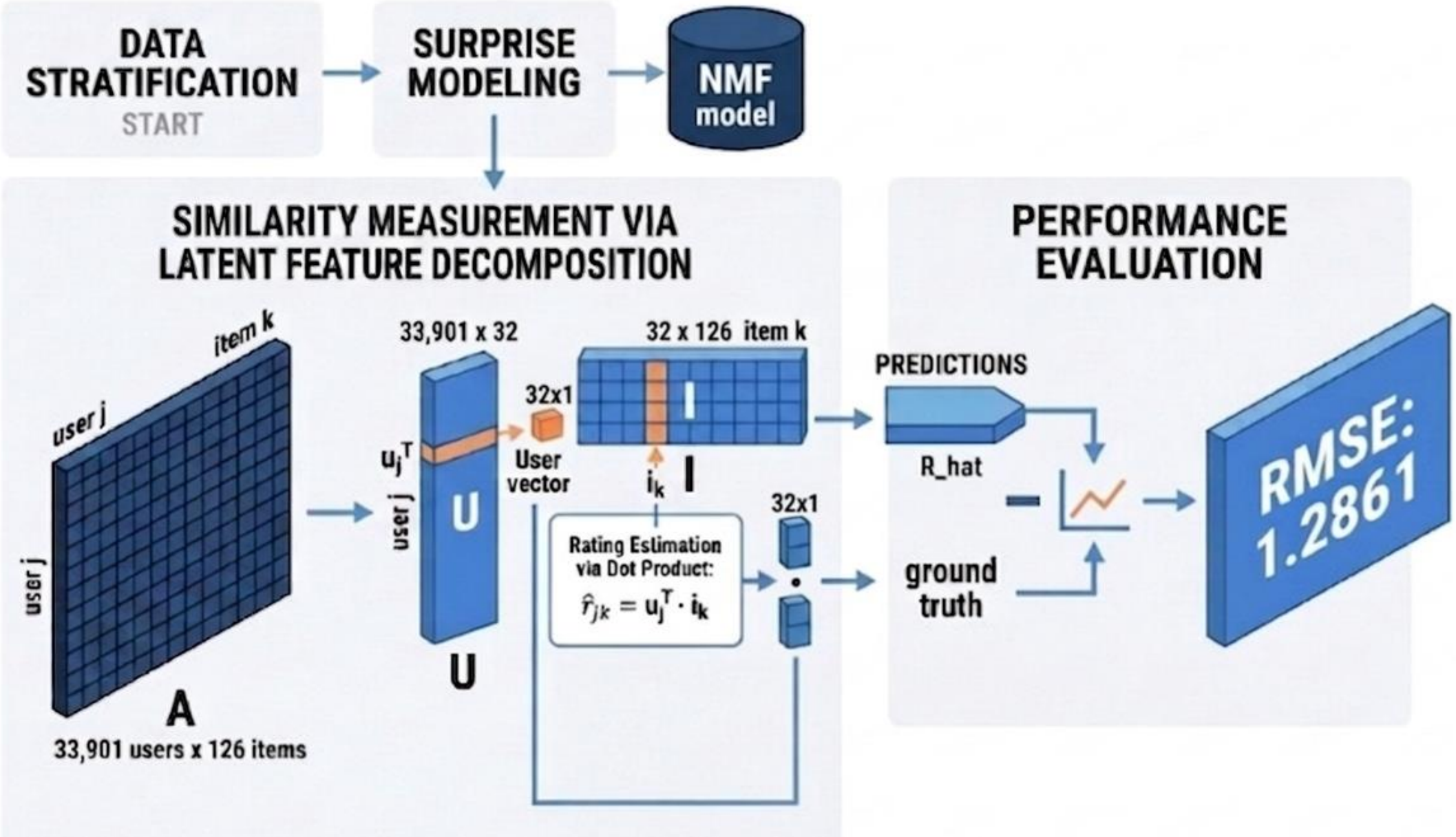
Collaborative-filtering Recommender System using Supervised Learning



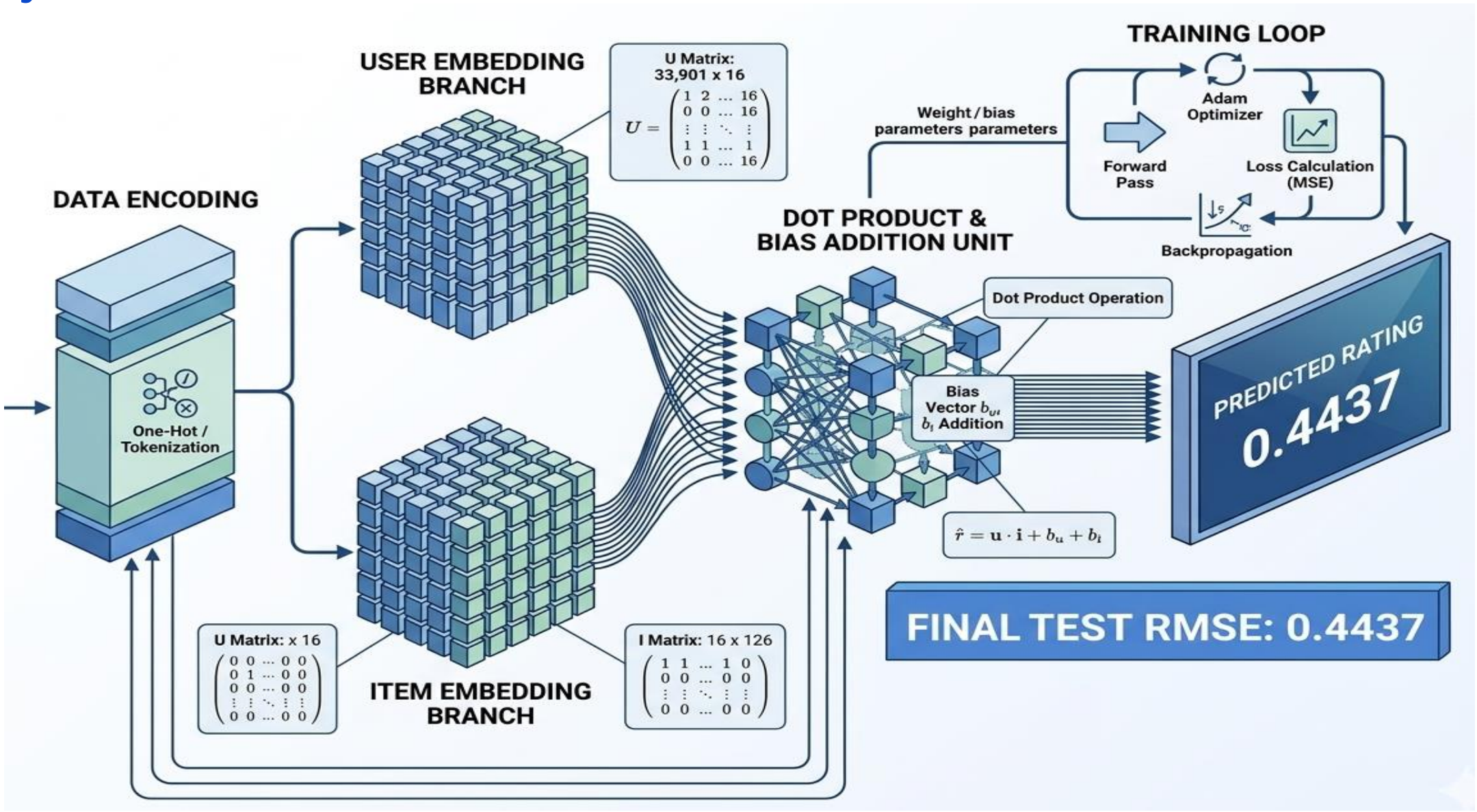
Flowchart of KNN based recommender system



Flowchart of NMF based recommender system

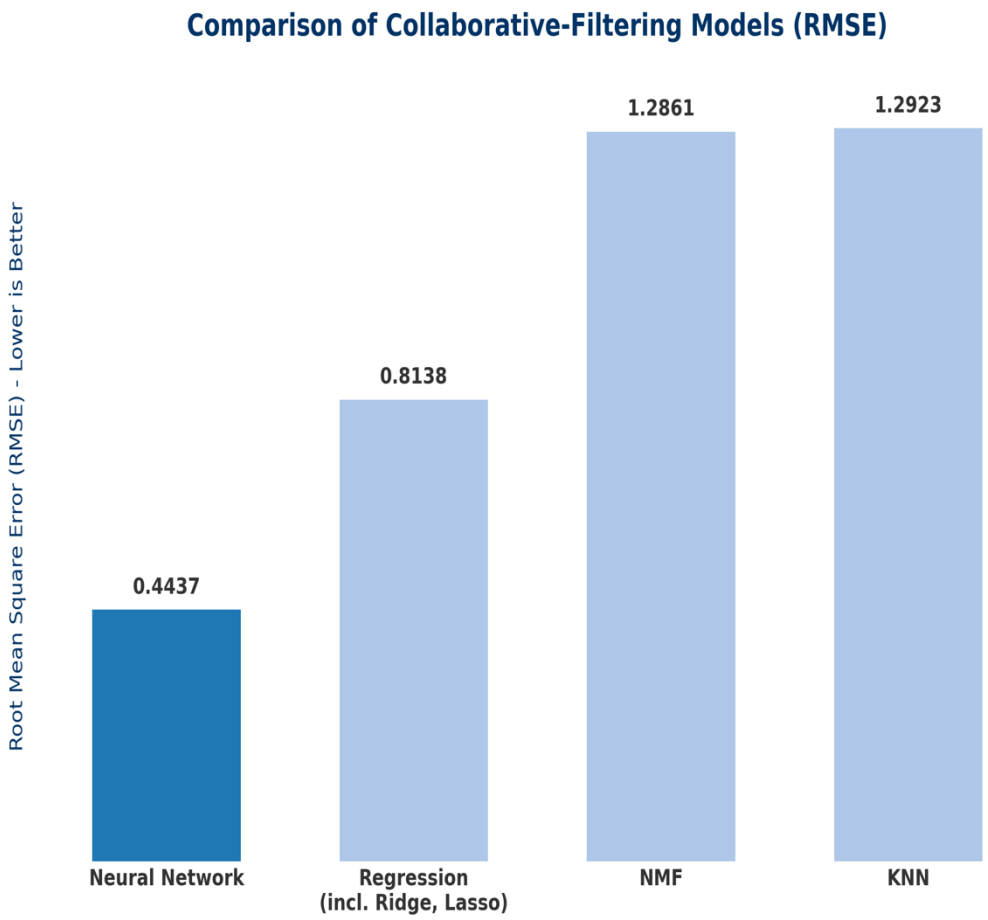


Flowchart of Neural Network Embedding based recommender system



Compare the performance of collaborative-filtering models

- **Neural Network (Winner):** Best performance with **RMSE = 0.4437**
- **Classification Attempt Failed:** Initial attempts at non-linear classification (RF/SVM) failed to beat a random-guess baseline (33.3%). This led to the pivot toward Neural Collaborative Filtering, which successfully captured the data's complexity.
- **Final Outcome:** Neural Network reduced prediction error by **~65% vs. KNN.**



Conclusions

- **Neural Networks as Gold Standard:** Best performance (**RMSE = 0.4437**), reducing prediction error by **~65% vs. KNN**
- **Classification Limitation:** Random Forest & SVM failed on sparse data, capped at **33.3% accuracy** (random guess level)
- **Reduced Choice Paralysis:** “Discovery Pool” narrows **126+ courses** into a personalized **Next Best Step** (e.g., DS0103EN)
- **Scalable Solution:** Streamlit deployment demonstrates **production-ready, real-time recommendation system**

Appendix

Component	Choice	Purpose
Embedding Size	16	Represents each User/Course as a 16-dimensional vector.
Activation	**ReLU**	Adds non-linearity, allowing the model to learn complex interest patterns.
Optimizer	**Adam** (lr=0.001)	An adaptive learning rate optimizer that helps the model converge faster.
Loss Function	**MSE**	Minimizes the squared difference between predicted and actual ratings.
Regularization	**L2** (1e-6)	Prevents "Overfitting," ensuring the model works on new, unseen students.
Initialization	**He Normal**	Statistically optimized starting weights for the neural layers.